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MILITARY SPECIFICATION

SEMITRAILER, COMPRESSED GAS CYLINDER: HELIUM, COMMERCIAL

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial semitrailer for storing, transporting, and charging compressed helium gas. The tandem axle semitrailer mounts 38 cylinders having a combined water capacity of not less than 340 cubic feet and a design working pressure of at least 2400 pounds per square inch gage (psig). Semitrailer procured under this specification are commercial items which shall be warranted as specified in the procurement documents.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

* ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2330

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MILITARY

MIL-V-2	- Valve, Cylinder, Gas (For Compressed or Liquefied Gases), General Specifications For.
MIL-T-5542	- Thread Compound, Antiseize and Sealing, Oxygen Systems.

STANDARDS

MILITARY

MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
MS75021	- Connector, Receptacle, Electrical -12 Contact, Intervehicular, 28-Volt, Waterproof.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Dangerous Articles in Tank Motor Vehicles.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Power Test Code

PTC 19.5:4	- Supplement on Instruments and Apparatus; Part 5, Chapter 4, Flow Measurement by Means of Standardized Nozzles and Orifice Plates.

Boiler and Pressure Vessel Code, Section VIII

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(Copies of the ASME Power Test Code may be obtained from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)
Pamphlet S-l Safety Relief Device Standards.

(Copies of the CGA pamphlet may be obtained from the Compressed Gas Association, Incorporated, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

* SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
* J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Fifth Wheel Kingpin, Commercial Trailers and Semitrailers (DoD adopted).
J702	- Brake and Electrical Connection Locations (DoD adopted).
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.2, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data except the general configuration shall be as shown in figure 1. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity prior to delivery of the items. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment, painting identification marking and data plates. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223.

3.1.1.1.1 Type gas. The name of the gas, "HELIUM, " shall be painted on each side of the semitrailer forward of the control cabinet assembly in block letters 4 inches high. The lettering "FOR USE WITH HELIUM ONLY" shall be painted on the inside and outside of each control cabinet assembly rear access door in block letters 2 inches high.

3.1.1.1.2 Control marking. All valves, gages, controls, and indicators used in operation of the semitrailer shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light and moisture will not fade or cause them to become eradicated.

3.1.1.2 Drain plugs. All drain plugs installed throughout the semitrailer for the purpose of draining lubricants shall be of the magnetic type.

3.2 Vehicle weight and ratings.

3.2.1 Curb weight. The curb weight of the semitrailer shall include all attachments, accessories, equipment and the 38 helium pressure vessels installed on the vehicle.

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3.2.2 Payload. The payload capacity shall be defined as the weight of 52,000 cubic feet of helium gas at 2,400 psig. For the purpose of computing the payload distribution on the suspension and kingpin it shall be considered as 1,000 pounds.

3.2.3 Gross weight. The gross weight of the semitrailer shall consist of the curb weight and payload.

3.2.4 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

3.3 Design. The semitrailer shall be designed to provide a readily mobile high-pressure gas storage pressure vessel assembly suitable for filling with helium gas at a generating site or railroad siding, transporting the gas to a using location, storing the helium gas, and charging the gas with the using apparatus at high flow rates. The semitrailer cylinder mountings and hold down devices shall be designed to withstand an acceleration factor of 1-1/2 g (g = gross weight of the semitrailer), for a minimum of 0.1 second in any and all directions. The complete system shall be capable of withstanding ambient temperatures up to plus 160 degrees Fahrenheit (°F) and exposure to solar radiation.

3.4 Performance. Performance shall be demonstrated with the loaded semitrailer equipped with all attachments, accessories, and the 38 pressure vessels; completely serviced and adjusted; and when being towed by a towing vehicle. The semitrailer shall be capable of being towed over improved roads at speeds up to 55 miles per hour (mph), and over dry improved gravel roads at speeds up to 10 mph with no evidence of failure, malfunction, or deformation of the semitrailer, components, attachments or accessories.

3.4.1 Tracking ability. The semitrailer shall follow the towing vehicle over dry, level, smooth, prepared, hard-surfaced roads and over dry improved and gravel roads without exceeding the tracking deviations specified in Federal Motor Carrier Safety Regulation 393.70(a). When coupled to the towing vehicle, operating in its minimum turning circle, the semitrailer shall follow without cramping or side slipping. The semitrailer shall be capable of assuming an angle of not less than 90 degrees with the longitudinal axis of the towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

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3.4.2 Service brakes. The service brakes shall control, decelerate and stop the semitrailer-tractor combination with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulations 393.45 through 393.52.

* 3.5 Dimensions and clearances. The minimum dimensions and clearances shall be as specified in table I, when measured with the semitrailer uncoupled from the towing vehicle, resting on the landing gear and tires, with the semitrailer horizontal to ground level, and without payload.

TABLE I. Dimensions and clearances.

Feature	Dimension
Overall length (feet) (max.)	30
Overall width (inches), (plus or minus 1 inch)	95
Upper fifth wheel height, from ground to underside of the upper fifth wheel plate (plus 1 inch, minus 0)	50
Distance from centerline of kingpin to front end of the semitrailer (plus 1 inch, minus 2 inches)	34-1/2
Swing radius from centerline of kingpin to most distant point on the semitrailer nose (not more than) (inches)	60
Landing wheel clearance (radius from centerline of the kingpin to any point 6 inches below the upper fifth wheel plate) (not less than) (inches)	82
Ground clearance, max. possible but not less than (inches)	14

3.6 Chassis components.

* 3.6.1 Chassis. Chassis components and vehicle ratings shall be the manufacturer's published ratings. Axle, suspension, wheel and tire ratings shall be for highway speeds (over 50 mph). Chassis construction shall be the manufacturer's standard frame or frameless type, but in either case the structural integrity requirements of DoT Safety Regulation 178.340 shall be met. Structural members shall support the running gear, upper fifth wheel plate and landing gear.

* 3.6.1.1 Frame. Semitrailer frame members shall be adequately braced with a sufficient number of crossmembers to insure stability. When the vehicle is loaded with specified payload (see 3.2.2), the maximum fiber stress in any frame member shall be not greater than 33 percent of the yield strength of the materials used. Under all conditions of operation, the frame members shall maintain chassis alinement without applying stress to the bulkheads, the cabinet or the

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pressure vessels. The outside frame members shall run the full length of the load space. The rear crossmember shall extend the full width of the semitrailer. All parts of the frame shall be completely electric-arc welded. Shear and moment diagrams with maximum deflection calculations for the semitrailer loaded shall accompany each questionnaire (commercial vehicle engineering data). Where frame members are spliced, joints shall be reinforced to fulfill the dynamic loading requirements and the static loading requirements. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

* 3.6.2 Rubber mounted suspension system. The semitrailer shall be furnished with a completely rubber mounted suspension system without radius rods, except that the spring mounting which compensates for horizontal movement need not be mounted in rubber. The rubber mounted suspension system, with components, shall have a rated capacity at least equal to the loads imposed upon each member, measured at the ground, with the semitrailer loaded with rated payload as specified in 3.2.2. When the suspension is rated at the spring pads, the unsprung weight shall be deducted. Clearances shall preclude interference between tires and any other part of the semitrailer under the operating conditions specified herein.

3.6.3 Tandem axle. Tandem axles shall be furnished. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground with the semitrailer loaded with rated payload as specified in 3.2.2. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for replenishing the oil supply. The hub caps shall incorporate provisions for venting or an equivalent method of withstanding pressures without leakage.

3.6.4 Wheels, rims, tires and tubes.

* 3.6.4.1 Wheels, rims and tires. The rim and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on the semitrailer. Tires shall be steel belted radial ply or, when specified (see 6.2), bias ply. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded to specified payload. When specified (see 6.2), disc type wheels shall be furnished. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

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* 3.6.4.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with Tire and Rim Association recommendations.

* 3.6.4.3 Carrier. A carrier for a spare wheel or rim and a tire shall be installed in a readily accessible location. Means shall be provided for securing the tire to the carrier to prevent accidental loss.

* 3.6.4.4 Spare wheel. When specified (see 6.2), a spare wheel or rim shall be mounted on the tire carrier.

* 3.6.4.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, load range and tread design as the tires installed on the semitrailer.

3.6.5 Brakes.

* 3.6.5.1 Brakes. The brakes shall be of the full air internal expanding type and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45, 393.46, and 393.47. The brake system shall include piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, air reservoirs with remote control drain, automatic slack adjusters, brake chambers, relay emergency valve, and all other parts and connections required for a complete air brake system. Gladhands shall conform to SAE J318. The location of the gladhands shall be as recommended in SAE J702. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier. No part of the brake system shall extend below the bottom of wheel rims to insure, in case of a flat tire, that the weight of the semitrailer will be supported by the rim and the flat tire and will not be imposed on any component of the braking system.

* 3.6.5.2 Parking brakes. Spring or air diaphragm mechanical lock type parking brakes shall be provided. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency braking conditions. The parking brakes shall remain in

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the applied condition with no additional application and despite the depletion of all air pressure. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.6.6 Fifth wheel. The upper fifth wheel plate shall be of the one piece, rigid type, not less than 1/4 inch thick, reinforced, having minimum dimensions of 42 inches long by 36 inches wide, and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The forward end of the upper fifth wheel plate shall have a turned-up lip for ease of coupling and for body protection. A separate turned-up lip plate may be furnished when the upper fifth wheel plate terminates at the forward end of the vehicle body. The turned-up lip plate shall be not less than 1/4 inch in thickness, and not less than 36 inches in width, and of a length to provide a turned-up lip for ease of coupling and for body protection. The plate shall be ground flush with the upper fifth wheel plate. A 2-inch kingpin, conforming to SAE J700, shall be provided. The kingpin shall be fabricated of heat treated alloy steel.

3.6.7 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating, landing legs, with two speed gears, and a handcrank on the curb side. Landing legs shall be provided with steel wheels or self-leveling skid pads. A holder shall be provided for the handcrank when not in use. The landing gear shall withstand, without permanent deformation, the combined static and dynamic forces due to the portion of the gross weight sustained, and the additional forces resulting from impact during coupling and uncoupling operations. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate at the kingpin from 46 inches to not less than 54 inches from the ground. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functioning or mechanical efficiency. The minimum ground clearance with the landing gear retracted shall be 14 inches.

3.6.8 Rear end protection. Semitrailer rear end protection shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.86. The bumper shall include a platform for the operator to stand on and to operate and service the controls in the cabinet.

3.6.9 Fenders. Fenders shall be furnished at the front and over the rear bogie. The fenders at the front shall protect the semitrailer from road splash from the wheels of the towing vehicle. Fenders at the rear of the semitrailer shall protect the rear cabinet and adjacent area from road splash. The fenders shall be constructed of sheet steel of not less than 14 gage (0.0747 inches). The fenders shall be flanged and reinforced.

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3.6.9.1 Rear wheel splash and stone throw protection. Rubber mud flaps to the rear of the rear wheels, in accordance with SAE J682, shall be provided.

* 3.6.9.2 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used.

3.7 Electrical equipment.

* 3.7.1 Lighting. The lighting system shall be of 12-volt potential and shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25, and 393.26. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected) or semitrailer bumper. The front of the semitrailer shall be equipped with a receptacle conforming to SAE J560 with the conductors connected and color coded as specified therein. The receptacle shall be located in accordance with SAE J702. Clearance lights shall be constructed for easy removal and replacement of lamps and lenses without the use of hand tools. Turn signal lamps shall conform to SAE J588.

* 3.7.2 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system independent of the 12-volt dc lighting system shall be provided for use with a towing vehicle equipped with a nominal 24-volt dc system. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.7.1. The 24-volt dc system shall conform to the Federal Motor Carrier Safety Regulations referenced herein, except the 24-volt dc stop and turn lamps shall be combined.

* 3.7.2.1 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt-dc system (see 3.7.1) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

* 3.7.2.2 Receptacle, 24-volt dc. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021, part number MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

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Contact B. Connect to left-hand turn signal and stop lamp (yellow).
Contact D. Connect to ground (white).
Contact E. Connect to clearance, side marker, identification and tail lamps (black and brown).
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The rest of the contacts shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

* 3.7.2.3 Resistors. When an interconnected 24-volt dc system is specified (see 3.7.2.1), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 28-volts dc to 12-volts dc. Each circuit resistor shall be selected to reduce the 28-volt dc regulated voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing, shall be provided with adequate ventilation or a heat sink to preclude overheating and any damage to resistors, wiring, or adjacent components.

3.8 Body. The body shall consist of front and rear bulkheads to retain the pressure vessels (cylinders), a rear valve control cabinet and cylinder supports transverse to the semitrailer.

3.8.1 Arrangement. The semitrailer shall consist of high-pressure gas cylinders mounted lengthwise on a semitrailer chassis with the cylinders manifolded together through individual shutoff valves so that any one cylinder, any combination of cylinders, or all cylinders simultaneously can be charged or discharged through the cylinder manifold assembly. The control valves shall be located in a cabinet at the rear of the unit. The semitrailer chassis and cylinder mountings shall be so designed that the cylinders will not form any part of the chassis structure but will be carried thereon as a nonstrength payload.

3.8.2 Cylinder mounting. The cylinders shall be mounted on the chassis to provide the minimum practicable overall vehicle width and the lowest practicable vehicle overall center of gravity.

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3.8.2.1 Support. The cylinder assembly shall rest on and shall be firmly secured to not less than four chassis cross-bearers located at approximately equal intervals along the straight cylindrical portions of the cylinders. The chassis cross-bearers shall be equipped with provisions for securely retaining the bottom horizontal row of cylinders in proper position when the semitrailer is subjected to the conditions specified herein.

3.8.2.1.1 Securing. The cylinder assembly shall be secured at each chassis cross-bearer with a strap or other arrangement capable of maintaining the assembly in compression and insuring against cylinder movement under all of the conditions specified herein. The strapping down equipment shall be such as to pull the cylinders downward and inward into the stack. Provisions shall be made for adjusting the tiedown strap tension and for its release without breaking a weld or seal in case of necessity for cylinder removal.

3.8.2.1.2 Separators. The cylinder assembly shall be designed and constructed to insure against metal-to-metal contact between the cylinders or between the cylinders and mounting and hold-down structures, except through the neck locknuts. Such metal-to-metal contact between the cylinders and between cylinders and other components shall be prevented by resilient nonmetallic separators interposed between the metal surfaces concerned.

3.8.2.1.2.1 Separator material. The nonmetallic separator material shall be resilient but shall not flow under pressure. Closely woven fabric material impregnated with an appropriate and properly cured rubber compound or with an appropriate rubber-like plastic, shall be considered a satisfactory spacer material. The separator material shall be of such thickness that when fully compressed under strapping down, cylinder, and charging loads, there will be ample space between the cylinders for drainage. A spacing of 1/4 inch between cylinders will be satisfactory. Dimensions of each spacer shall be such as to insure that pressure on the spacer due to cylinder weight, strapping down forces, charging load, and acceleration forces will be sufficiently low to insure against exceeding the elastic limit of the material or causing the material to flow.

3.8.2.2 Bulkheads. A steel bulkhead containing holes through which the threaded cylinder necks protrude, and to which the cylinders can be secured by locknuts on the cylinder neck external threads, shall be provided at each end of the cylinder assembly. The holes in the bulkheads shall be so spaced and of such size as to permit secure fastening of the cylinders to the bulkheads with the locknuts and to prevent the bulkheads acting as a support for any cylinder. Locknuts shall be provided on each side of the bulkhead.

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3.8.2.2.1 Size. The bulkheads shall be of sufficient overall width and height to extend not less than 2 inches beyond the extreme cylinder outline as viewed from the end. Bulkhead thickness shall be sufficient to aline the cylinders in the proper longitudinal position and to insure against longitudinal movement of the cylinders when the semitrailer is subjected to the acceleration forces specified herein (see 3.3).

3.8.2.2.2 Installation. Bulkhead bottoms shall be secured to the chassis, and the tops shall be braced by compression or tension members so that longitudinal movement of the cylinders will not occur when the semitrailer is subjected to the maximum acceleration forces specified herein (see 3.3). Bulkheads shall be installed to be readily demountable for removal and replacement without the use of special tools and without breaking welds or seals.

3.8.3 Cylinder valves. The rear end of each cylinder shall be provided with a diaphragm-type, oil-free, helium cylinder valve designed for service on cylinders with 2,400 psi working pressure and conforming to MIL-V-2 except for the safety device.

* 3.8.4 Cylinder safety equipment. Both ends of each cylinder of the storage cylinder assembly shall be equipped with Type CG-5 (frangible disc) safety relief devices conforming to CGA Pamphlet S-1. The safety relief devices shall release the cylinder contents when the cylinder pressure exceeds 3,800 ⇦ 200 psi at ⇦ 71 degrees Celsius (ꝏC) and shall include a means of piping away any gas released, to a point above the highest tube in the trailer and away from personnel on the ground (see 3.9.4.10). The safety device at the rear end of each cylinder (rear end of the semitrailer) shall be constructed integral with the cylinder valve. The safety device at the front end of each cylinder shall be installed so that it will be protected against damage by impact.

3.8.4.1 Volume calibration. Each cylinder shall be marked with a serial number and calibrated for actual (water) volume in cubic inches at pressures of 1,000, 2,000, and 2,500 psi and at a temperature of 20°C plus or minus 5°C. Each cylinder shall be permanently and legibly stamped with the actual (water) volume at each of the specified pressures. The volume stamps shall be placed near the cylinder serial number where both will be simultaneously visible.

* 3.8.4.2 Temperature indicators. Cylinder(s) shall be equipped with an accurate, minimum-lag sensing element capable of measuring the average temperature of the gas

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contained in the cylinder. All cylinder(s) temperature sensing elements shall be wired to a single temperature indicator through a selector switch designed to connect the indicator to any selected individual sensing element.

3.8.4.2.1 Mounting. The temperature indicator and selector switch shall be securely mounted in the control cabinet adjacent to the manifold pressure gages specified in 3.9.4.3. They shall be readily visible and accessible to an operator standing on the ground at the rear of the semitrailer. The mounting shall be such as to protect the units against damage during service and road transport, and to permit removing the units without disturbing components not associated with the temperature-measuring apparatus.

* 3.8.5 Cylinders. The vehicle shall be equipped with 38 cylinders which have a total water capacity of not less than 340 cubic feet. The nominal water capacity of each cylinder shall be at least 15,460 cubic inches. The unit, including all operating equipment, shall conform to Title 49, Code of Federal Regulation 173.300, 173.302, 173.306, 173.314 and the applicable portions of Section 178 requirements specified herein. The cylinders shall conform to United States Department of Transportation specification 3AA-2400. The cylinders and all associated equipment shall be constructed to withstand 2,400 psig working pressure, and all cylinders and connections thereto shall be designed and fabricated in accordance with Title 49, Code of Federal Regulations, Parts 1 through 199, and ASME Boiler and Pressure Vessel Code Section VIII, and shall be stamped with the Code 4-symbol.

3.8.5.1 Pressure relief devices. Wherever needed, pressure relief devices conforming to Compressed Gas Association (CGA) Pamphlet 8-1 shall be provided to protect against excessive pressure in any component. Provisions shall be incorporated to permit discharge of pressure without danger to operating or maintenance personnel.

* 3.8.5.2 Storage capacity. The semitrailer shall have a total actual (water volume) gas storage cylinder capacity of not less than 52,000 cubic feet at 2,400 psig.

3.8.5.3 Working pressure. The semitrailer gas storage cylinders and all other gas-handling components shall be capable of containing the gas without leakage at a working pressure of 2,400 psi and at any temperature between minus 62°C and plus 71°C.

3.8.5.4 Storage cylinder assembly. The storage cylinder assembly shall consist of 38 cylinders constructed in accordance with Department of Transportation Specification 3AA-2400 for a working pressure of not less than 2,400 psi under the temperature and use conditions

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specified herein. The cylinders shall be necked down and threaded both internally and externally on each end.

* 3.8.5.5 Cylinder dimensions. Each cylinder shall be not more than 22 feet in length nor more than 10 inches in outside diameter. The external neck threads shall be not less than 2-1/8 inches in length nor less than 2 inches in diameter. The internal neck threads shall provide for installation of type 07-581-1 oil-free helium valves conforming to MIL-V-2.

3.8.5.6 Servicing rate. The semitrailer valving and manifold system shall be of sufficient gas-handling capacity to insure that, when all individual cylinder shutoff valves are open to the manifold and all cylinders are at a pressure not exceeding 250 pounds per square inch gage (psig), the semitrailer will be capable of delivering helium gas through the manifold final outlet fitting against atmospheric pressure at a rate of not less than 1,000 cubic feet per minute at 760 millimeters mercury (mm Hg) and 21°C.

3.8.6.1 Control cabinet and assembly. A sturdy, sheet-metal cabinet shall be provided for the cylinder valves, cylinder manifold assembly, and semitrailer accessories. The cabinet shall be mounted on the rear of the semitrailer chassis with the rear bulkhead forming the forward wall.

3.8.6 Cabinet.

3.8.6.1 Control cabinet and assembly. A sturdy, sheet-metal cabinet shall be provided for the cylinder valves, cylinder manifold assembly, and semitrailer accessories. The cabinet shall be mounted on the rear of the semitrailer chassis with the rear bulkhead forming the forward wall.

3.8.6.1.1 Cabinet design. The cabinet assembly shall be designed and constructed to protect the enclosed components from mechanical damage, direct solar radiation, rain, snow, sleet, sand and mud. The top of the control cabinet and the tops of all protruberances therefrom shall be shaped and slanted to resist the retention of rain, snow, or sleet. The operating control panels shall be constructed so that snow or ice will not lodge thereon and prevent easy access or operation.

3.8.6.1.2 Removal. The cabinet shall be constructed so that it can be readily removed without disturbing the storage cylinders or cylinder manifold assemblies.

3.8.6.1.3 Ventilation openings. Ventilation openings shall be provided at or near the top of the control cabinet to prevent accumulation of gases in case of a leak. The ventilation openings shall be so located or shielded that the entrance of rain, snow, or other precipitation will be prevented.

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3.8.6.1.4 Size. The cabinet shall be of sufficient size to provide ready access to enclosed components for operation, maintenance, and repair. The cabinet shall be of full-flanged, flush type construction.

* 3.8.6.1.5 Access doors. Twin, full-length, vertically hinged, full-flanged, flush type, swinging doors shall be provided. The doors shall open from the control cabinet center (hinged at the body sides) with no center post, to provide an unrestricted opening into the control cabinet. The width of the doors shall be sufficient to provide free access to all cylinder valves and all cylinder manifold assembly components when the doors are open. Means shall be provided for securing the doors in the open position. Unless manually actuated, the latches shall not release the doors when the semitrailer is subjected to wind gusts. Access door latches shall be corrosion resistant and shall be constructed to permit operation by personnel wearing heavy mittens. They shall include provisions for looking with common padlocks.

3.8.6.2 Cabinet lighting. Not less than two 12-volt dome lamps shall be installed in the control cabinet to provide sufficient lighting for safe operation of the trailer. The dome lamps shall be designed to operate on electric energy supplied from the towing vehicle. Light switches shall be provided in the cabinet to control the lights manually and to turn off the lights when the cabinet doors are closed.

3.8.6.3 Portable lighting.

3.8.6.3.1 Floodlamp sockets. Disconnect sockets for attachment of the floodlamps specified in 3.8.6.3.2 shall be securely mounted in a convenient location at the right and left corners of the control cabinet. The floodlamp sockets shall be wired into the dome lamp circuit to be always on.

3.8.6.3.2 Floodlamp. A weatherproof, quickly attachable, portable floodlamp, equipped with a bulb not less than 50 candle power (cp), shall be provided with each semitrailer. The floodlamp shall be designed to operate on 12-volt electric energy supplied from the truck tractor through the disconnect sockets specified in 3.8.6.3.1.

3.8.6.3.3 Attachment device. The floodlamp shall be provided with a quick-attachment device with which it may be quickly and readily attached to either rear corner of the semitrailer in a convenient location where its light will not be blocked by panels or doors. The quick-attachment device shall include a ball-and-socket or similar joint that will permit the light to be directed anywhere in a 270 degree arc around the corner of the corner of the semitrailer to which it is attached.

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3.8.6.3.4 Extension cord. The floodlamp shall be provided with a heavy duty flexible extension cord not less than 15 feet in length. One end of the cord shall be permanently attached to the lamp. The other end shall be equipped with a disconnect plug connected to fit the disconnect socket specified in 3.8.6.3.1.

3.8.6.3.5 Switch. A manually operated on-off switch shall be provided either on the lamp body or in the cord adjacent to the lamp.

3.8.6.3.6 Storage provisions. Facilities shall be provided within the control cabinet assembly for protection against damage and for storage of the floodlamp.

3.8.6.4 Handbook compartment. A handbook compartment shall be provided in the control cabinet assembly at a location readily accessible to operating and maintenance personnel.

3.8.6.5 Padlocks. Heavy duty corrosion resistant padlocks shall be provided for the control cabinet access door latches and the tool compartment. Not less than three master keys shall be provided with each semitrailer.

3.9 Cabinet manifold, valves and gages.

3.9.1 Cylinder manifold assembly. The outlets of all individual cylinder valves shall be manifolded together for discharging or filling through two valve-controlled final outlets located in the center rear of the semitrailer. The manifold assembly shall consist principally of the manifold piping, pressure gage(s), two master flow control and shutoff valves, and two final outlet fittings.

3.9.2 Design and construction. The manifold assembly shall be designed and constructed to handle helium gas without leakage during cylinder filling or discharging operations and under the mobile operating and environmental conditions specified herein.

3.9.3 Manifold materials. The manifold assembly shall be constructed of corrosion-resistant material with high impact resistance, such as copper or brass. Manifold piping, fittings, and attachments shall be fabricated from materials that are chemically compatible with each other or basically the same type material.

3.9.4 Manifold piping. The manifold piping shall consist of the manifold header plus a branch line between each cylinder valve outlet and the header. All manifold piping shall be of sufficient inside diameter to insure compliance with the delivery flow rate requirement specified in 3.8.5.6.

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3.9.4.1 Manifold header. The manifold header shall be fabricated from rigid, seamless pipe or tubing of not less than 3/4-inch inside diameter. Tubing wall thickness shall be sufficient to provide a safety factor of not less than 4 and shall resist damage in normal handling. The manifold header shall be provided with connections for each of the cylinder branch lines, pressure gage(s), two master flow control and shutoff valves, and two final outlet fittings.

3.9.4.2 Branch. Branch lines between the manifold headers and the individual cylinder valve outlet connections shall be constructed from semiflexible, seamless tubing of sufficient inside diameter for insuring compliance with the flow requirements specified in 3.8.5.6. Tubing wall thickness shall be sufficient to provide a safety factor of not less than 4 and resist damage in normal handling. Each branch line shall be fabricated with a loop or curvature that will insure against placing a strain on the manifold piping in case of relative movement between the cylinders and the manifold assembly.

* 3.9.4.3 Manifold pressure gages. Manifold pressure gage(s) shall be provided for indicating manifold and servicing line pressures. Each pressure gage shall have a dial diameter of not less than 4 inches, shall be graduated over approximately 270 degrees of the dial face, and shall be accurate to within 1 percent of full-scale range. The gage(s) shall have safety backs.

* 3.9.4.3.1 Range. The range of the pressure gages shall be 0 to 5,000 psi. The gages shall be graduated with subdivisions not greater than 25 psi.

* 3.9.4.3.2 Mounting and connections. The pressure gage(s) shall be securely mounted to a panel, frame members, or brackets near the master flow control and shutoff valves in positions readily visible to an operator standing on the ground at the rear of the semitrailer. The pressure gage(s) shall be attached to the manifold pressure sensing connections by means of semiflexible, seamless tubing. The tubing or connections shall not be used for or be a part of the gage mounting. The mounting shall protect the gage(s) against damage during service and road transport, and shall permit removing the gage(s) without disturbing or disconnecting other components.

3.9.4.3.2.1 Connection locations. One pressure sensing connection shall be located in the manifold header between the branch lines and the master flow control and shutoff valves. The other sensing connections shall be located between the master flow control and shutoff valves and the final outlet fittings.

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3.9.4.3.3 Shutoff valves. Each pressure gage sensing connection shall incorporate a readily operable and accessible shutoff valve capable of isolating the gage from the manifold assembly.

3.9.4.4 Master flow control and shutoff valves. The master flow control and shutoff devices shall be manually operated valves capable of throttling and controlling the flow of 2,400 psig helium. The valves shall be of sufficient size to insure compliance with the delivery flow rate requirements specified in 3.8.5.6 but shall be not less than 3/4 inch pipe size. Construction of the valves shall be such that they will not cause unnecessary restriction of the line when open and will isolate the final outlet fittings from the remainder of the manifold piping when closed. The valves shall be readily operable by hand. The valves shall be mounted and installed so that the control handles will be readily accessible to an operator standing on the ground at the rear of the semitrailer. Force exerted in valve opening or closing shall not be transmitted to the manifold piping.

3.9.4.5 Final outlets. Each cylinder manifold final outlet shall be the hose disconnect coupling half assembly and shall be 3/4 inch. The manifold piping shall terminate in fittings provided with threads for direct and leak-tight installation of the specified disconnect coupling half assemblies.

3.9.4.5.1 Outlet mountings. The final outlet fittings shall be mounted and installed to be readily accessible to an operator standing on the ground at the rear of the semitrailer for attachment of the servicing and charging hoses specified in 3.9.5. Each final outlet fitting shall be so mounted to supporting members that force applied in attaching or removing the servicing hoses, or while a servicing hose is attached, will not be applied to the manifold piping.

3.9.4.5.2 Protective caps. Protective caps designed to tightly seal the fittings and provide mechanical protection for the threads shall be provided. The caps shall be securely attached to the final outlet fittings by chains or similar devices to prevent loss.

* 3.9.4.6 Cylinder manifold vent attachment. A valve-controlled vent line, fabricated from pipe or tubing not less than 3/4 inch in size, shall be provided between the cylinder and venting manifold assemblies to permit emergency release of contents through the venting manifold assembly.

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3.9.4.6.1 Vent valve. The vent control shall be a manually operated shut off valve of not less than 3/4 inch in size designed for the control of 2,400 psi helium gas. Construction of the valve shall be such that it will not cause unnecessary restriction of flow when open and will isolate the cylinder manifold assembly from the venting manifold assembly when closed. The valve shall be readily operable by hand.

3.9.4.6.1.1 Valve mounting. The vent control valve shall be mounted with the cylinder manifold assembly in a position readily accessible to an operator standing on the ground at the rear of the semitrailer. Force exerted in valve opening or closing shall not be transmitted to the manifold piping.

3.9.4.7 Connections. Detachable connections shall be provided between the branch lines and individual cylinder valve outlet fittings, between the manifold header and pressure gages, between the manifold and master flow control and shutoff valves, and between the master flow control and shutoff valves and the final outlet fittings. All other connections between manifold parts and fittings shall be permanently sealed by soldering, welding, or other means, unless disconnecting is necessary for manifold removal, removal of other components, maintenance, or servicing.

3.9.4.8 Cylinder manifold assembly installation. The cylinder manifold assembly shall be installed within the control cabinet at the rear of the semitrailer. All valves and controls shall be installed and located to be readily accessible to an operator standing on the ground at the rear of the semitrailer, or on the step specified in 3.6.8.

3.9.4.8.1 Manifold mounting. The manifold assembly shall be securely mounted to the semitrailer in a manner that will insure against dependence on connections for support and that will permit removal of the manifold assembly as a unit without bending, springing, or otherwise damaging lines or connections. Mounting shall be sufficiently solid to insure against damage when the semitrailer is towed over off-roads or subjected to the maximum acceleration forces specified herein (see 3.3).

3.9.4.9 Venting manifold assembly. A manifold assembly shall be provided for disposing of vented gas through vent connections external to any confined space of the semitrailer.

3.9.4.10 Design and construction. The venting manifold assembly shall be designed and constructed to receive and dispose of all gas released by the safety devices specified in 3.8.4 or manually vented during transport. All parts and connections of the venting manifold shall be

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designed for a working pressure of not less than 2,400 psi. The venting manifold assembly shall be fabricated from basically the same material as the cylinder manifold.

3.9.4.10.1 Component parts. The venting manifold assembly shall consist of a manifold header at each end of the semitrailer, branch lines for connecting the individual cylinder safety relief device outlets to the headers, and vent fittings.

3.9.4.11 Manifold headers. A manifold header shall be securely installed across the semitrailer at each end of the cylinder assembly in a protected location. Each manifold header shall be provided with connections for attachment of individual branch lines from the safety devices on each cylinder plus connections for attachment of the header interconnecting piping. The rear header shall also include a connection for attachment of the cylinder manifold vent provision specified in 3.9.4.6. Manifold headers shall be fabricated from pipe or tubing of not less than 3/4-inch pipe size designed for the working pressure specified in 3.9.4.10.

3.9.4.11.1 Header connectors. The two manifold headers shall be interconnected by two sections of the same size and type piping from which the headers are fabricated. The connecting piping shall be so installed that, if the vent outlet fittings on one header are closed, any gas vented into that header will pass through the header connectors to be discharged from the vent outlets on the header at the other end of the semitrailer. The connecting piping shall be securely installed in protected locations within the storage cylinder assembly.

3.9.4.12 Branch line. A branch line with an inside diameter equal to or greater than those of the cylinder manifold assembly branch lines specified in 3.9.4.2 shall be attached between the header and each of the safety relief devices specified in 3.8.4. Attachment of the branch line to each safety relief device shall be such as to insure that all gas vented through the safety relief device will be delivered into the header.

* 3.9.4.13 Outlet fittings. Each end of each manifold header shall have an outlet fitting extending sufficiently outside the semitrailer enclosures to permit ready attachment of a venting hose. The fittings shall each be equipped with a 3/4-inch external pipe thread and a weathertight, low-pressure, blowout-type relief device that will vent the manifold at an internal positive pressure not to exceed 150 psig. A high pressure removable cap that can be readily installed when no venting hose is attached shall be provided for each outlet fitting. The high-pressure

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removable cap shall be designed to seal the outlet against leakage at an internal pressure of less than 3,000 ± 150 psi and to relieve the venting manifold to the surrounding space when this pressure is exceeded.

3.9.4.13.1 Outlet mounting. Mounting and installation of the outlet fittings shall be such that they are readily accessible. Force exerted in attachment, use, or removal of caps, vent lines, and other such attachments, shall not be transmitted to the manifold piping.

3.9.4.14 Detachable fittings. Detachable fittings shall be provided between the branch lines and the individual safety relief device outlet fittings, between the header connections and the headers, and between the cylinder manifold vent attachment and the rear header. Other connections shall be permanently sealed by soldering, welding, or other suitable means, unless disconnecting is necessary for manifold removal, removal of other components, maintenance, or servicing.

3.9.5 Servicing hose. A 100-foot long, high-pressure servicing hose shall be provided. The servicing hose shall be constructed for handling helium in filling the semitrailer gas storage cylinders or servicing external containers from the semitrailer. The hose shall be sufficiently flexible to permit easy handling, prevent damage to the hose from continued coiling and uncoiling, and permit coiling to a diameter convenient for ready storage within the control cabinet assembly. The weight of the hose shall be the minimum practicable consistent with the requirements specified herein. The inside diameter of the hose shall be not less than 3/4 inch. The servicing hose shall not leak when subjected to an internal pressure of 10,000 psi, and shall have a burst pressure of not less than 12,500 psi. The servicing hose outer covering shall be resistant to abrasions, acids, hydrocarbon fuels, greases, alkalis, and heat.

3.9.5.1 Hose connections. Each end of the servicing hose shall be equipped with a copper or brass hose disconnect coupling half assembly designed for quick and leaktight attachment to, and quick disconnecting from, the final outlets specified in 3.9.4.5. The hose disconnect coupling half assemblies shall have an inside free passage diameter equal to or greater than that of the final outlet disconnect coupling half assemblies. The disconnect coupling half assemblies shall be attached to the hose by threaded fittings to permit ready replacement.

3.9.5.2 Hose-to-pipe-thread adapter. The semitrailer shall be equipped with a hose disconnect coupling half assembly identical to the fitting specified in 3.9.4.5 for adapting the servicing hose to other equipment. If the disconnect coupling half assembly is provided with

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other than a national pipe thread, an additional adapter capable of withstanding a working pressure of 2,400 psi shall be supplied to provide for leaktight attachment of the disconnect coupling half assembly to a national pipe thread.

3.9.5.2.1 Hose-to-hose adapter. The semitrailer shall be equipped with an adapter designed for attaching two servicing hoses end-to-end to form a 100-foot hose when remote servicing of equipment is necessary.

3.9.5.2.2 Outlet-to-pipe thread adapter. The semitrailer shall be equipped with a hose disconnect coupling half assembly identical to those supplied on the servicing hose.

3.9.5.2.3 Charging adapter. The semitrailer shall be equipped with a charging adapter consisting of a hose disconnect coupling half assembly identical to those supplied on the servicing hose and a male connector compatible with model 3,000, 3/8-inch, Pioneer hydraulic coupling female connector, or equal. The adapter shall be so assembled that when the hose disconnect coupling half assembly is connected to one of the cylinder manifold assembly final outlets, a servicing hose equipped with a model 3,000, 3/8-inch, Pioneer hydraulic coupling female connector, or equal, can be readily attached and utilized for filling the semitrailer.

3.9.5.3 Tools. The servicing hose connections and the adapters shall be designed and constructed so that they can be readily attached or removed without the use of special tools.

3.9.5.4 Protective caps and plugs. Protective caps and plugs shall be provided for tightly closing and protecting the threads on all fittings when disconnected and not in use. Each cap or plug shall be securely attached to the fitting by a corrosion resistant chain or similar device to prevent loss.

3.9.6 Pipe thread compound. Pipe thread compound conforming to MIL-T-5542 shall be applied to all pipe thread fittings before assembly. The compound shall be capable of sealing threaded fittings against leaks of high-pressure helium gas.

3.10 Tool compartment. A compartment for storage of all tools required in normal operation and maintenance of the semitrailer shall be provided. Provisions for locking the tool compartment with a common padlock shall be incorporated. If the tool compartment is external to the control cabinet assembly, it shall be designed to prevent the entry of rain, road splash, and other foreign material.

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3.11 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each semitrailer for operational use, including at least the following: adjustment of electrical and brake systems; inflation of all tires; and complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.12 Finishes and protective coatings.

3.12.1 Exposed parts and surfaces. All exposed parts and surfaces of the semitrailer, with the exception of certain corrosion-resistant materials installed in the control cabinet assembly, the servicing hose, and corrosion-resistant disconnect fittings, shall be given protective coatings as specified herein (see 3.1.1.1).

3.12.1.1 Corrosion-resistant metal parts. Metal parts and surfaces installed within the control cabinet assembly and fabricated from corrosion-resistant material need not be given protective coatings if such coatings will interfere with their proper functioning.

3.13 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the best commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

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4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine net weight and distribution of net weight on kingpin and rear axles. The imposed loading on kingpin and rear axles will be computed using the net weight and the payload (see 3.2.2) distributed in the cylinders. The calculated imposed loads on rear axles will be utilized to ascertain that the suspension, axles, and tires are of adequate capacity to meet contractual requirements.

4.3.2 Road test. The first production vehicle shall be coupled to a truck tractor and shall be road tested with rated payload for a distance of not less than 200 miles. The vehicle shall be examined and tested to determine conformance to performance requirements specified in 3.4 through 3.4.2.

4.3.2.1 Road test preparation.

4.3.2.1.1 Preparation. The semitrailer shall be filled with dry, oil-free gas to a pressure of not less than 2,400 psi. The pressure and temperature of the gas within the cylinders shall be recorded, after which all valves shall remain closed throughout the road test. No servicing or adjustments will be permitted during the road test.

4.3.2.1.2 Road test procedure. The semitrailer shall be towed for 180 miles over improved roads at speeds up to 55 mph, and 20 miles over dry, improved gravel roads at speeds up to 10 mph.

4.3.2.1.3 Brake test. A minimum of five sudden stops, in which the semitrailer and towing tractor are brought to a complete halt from a speed of not less than 20 mph within a distance of 40 feet, shall be conducted during the improved road operation.

4.3.2.2 Road inspection. Throughout the road handling test, the semitrailer shall be observed as to its trackability, ease of handling, tendency to yaw and sway. Upon completion of the road handling test, the semitrailer shall be inspected for damage, misalinement, distortion, leaks, or any other defects or damage resulting from the testing. Temperature and pressure of the gas in the cylinders shall be recorded and compared with the data recorded under 4.3.2.1.2, prior

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to the start of the road test. Undesirable roadability features or any damage resulting from the test that would affect the semitrailer use and operation shall be cause for rejection.

4.3.2.2.1 Valve operation test. Each semitrailer valve shall be manually operated through its complete adjustment range and tightly closed not less than 10 times. The valve shall then be closed hand tight and subjected to a helium gas pressure of not less than 2,600 psi at the inlet. Leakage detectable by a soap film test (film across the outlet, between stem and body, and over all outside surfaces of the valve) shall be cause for rejection of the valve.

* 4.3.3 Safety device tests. Safety devices shall be tested as specified in CGA Pamphlet S-1 (see 3.8.5.1). The manufacturer shall submit to the engineering offices of the procuring activity a certification that the safety devices meet the requirements specified herein.

* 4.3.4 Hydrostatic test. The cylinder shall be hydrostatically tested in accordance with Title 49, Code of Federal Regulations, United States Department of Transportation specification 3AA. Distortion beyond the tolerance specified, damage or leaks resulting from the hydrostatic testing of each cylinder shall be cause for rejection of the cylinders. The manufacturer shall submit to the engineering offices of the procuring activity a certification that the hydrostatic test meets the requirements specified herein.

* 4.3.5 Gas pressure test. The servicing hose shall be attached to one manifold final outlet fitting; the hose-to-pipe thread adapter specified in 3.9.5.2 attached to the other end of the hose; all cylinder valves, the master flow control and shutoff valve controlling the outlet to which the hose is attached, and the cylinder manifold vent valve opened wide; and the venting manifold assembly outlet fittings sealed with the high-pressure removable caps specified in 3.9.4.13. The other master flow control and shutoff valve shall be closed. The storage cylinders, cylinder shutoff valves, fittings, cylinder manifold assembly, vent manifold assembly, servicing hose, and servicing hose disconnect fittings shall then be pressurized with dry, oil-free gas through the servicing hose to not let less than 2,600 psi and held at this test pressure for not less than 30 minutes. Compressed air may be used for this test if desired. Distortion, damage, or leaks resulting from this test shall be cause for rejection of the components affected. The manufacturer shall submit to the engineering offices of the procuring activity a certification that the gas pressure teat meets the requirements specified herein.

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* 4.3.6 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished (see 3.7.2.1), the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test the resistor assembly, including resistors, wiring and adjacent components, shall be examined for evidence of overheating, deterioration or damage.

* 4.3.7 Availability of documents. The contractor shall make available to the Government at the point of final acceptance a current copy of the documents listed in 2.2. The documents shall be surrendered to the Government for the personal use of the Government representatives in conjunction with the contract for the duration of the contract.

4.3.8 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The manufacturer shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.9 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of the vehicle until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. All semitrailers under contract shall be tested as specified in 4.3.3 through 4.3.6. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.11 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The semitrailers covered by this specification are intended for general non-tactical use by the Government in transporting and dispensing compressed helium gas from a generating or receiving point to the using site.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Bias ply tires, if required (see 3.6.4.1).
(d) Disc type wheels, if required (see 3.6.4.1).
(e) Spare wheel or rim, if required (see 3.6.4.4).
(f) A spare tire, if required (see 3.6.4.5).
(g) An independent 24-volt system, if required (see 3.7.2).
(h) An interconnected 24-volt system, if required (see 3.7.2.1).

6.3 Identification of changes. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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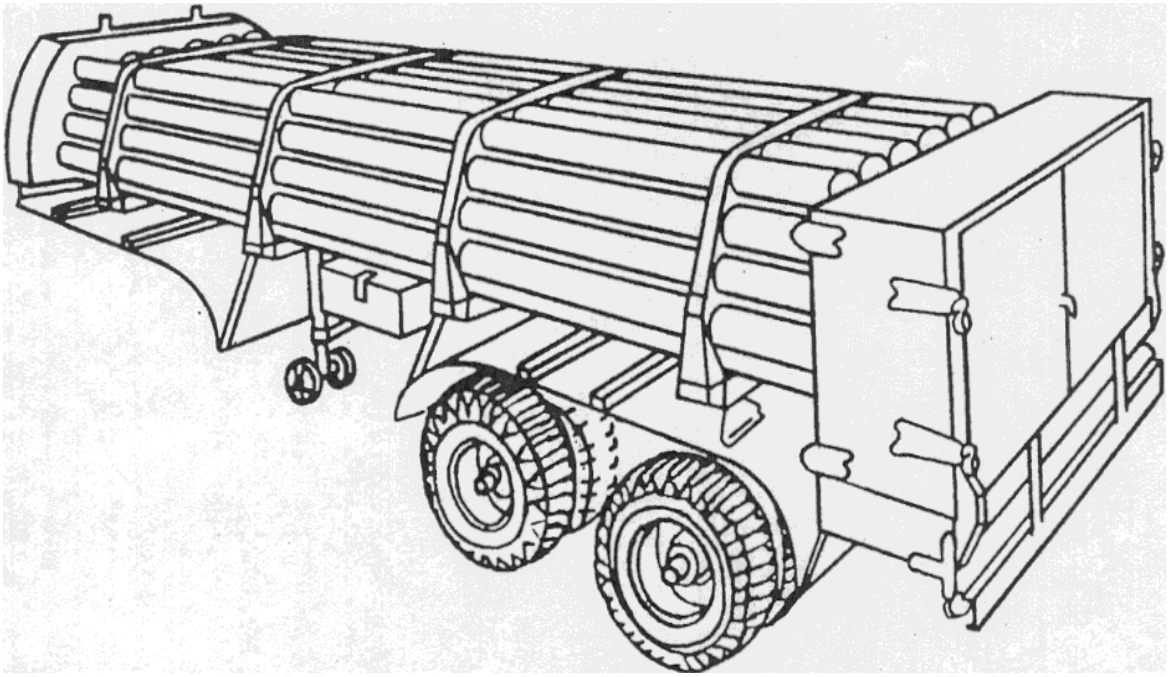


FIGURE I. SEMITRAILER, COMPRESSED GAS CYLINDER

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